Warming Up Cold-Start CTR Prediction with Adaptive Item-Specific Graph Diffusion

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1 Application Domain and Problem Formulation

1.1 CTR Prediction in Recommender Systems

Recommender systems rank items according to their relevance to users. Click-through rate (CTR) prediction estimates the probability that a user clicks an item after it is displayed. Accurate CTR prediction improves ranking, exposure allocation, user experience, and platform revenue.

Modern CTR models learn user–item feature interactions from historical data. However, their performance decreases for newly introduced items because these items have few or no interactions. This project therefore studies CTR prediction for new items and aims to provide reliable predictions as they gradually collect user feedback.

1.2 Cold-Start and Warm-Start Settings

In the cold-start setting, a new item has no historical interactions. The model must rely on item features and knowledge transferred from old items. This is difficult because the new item ID and its feature-interaction patterns have not been observed during training.

In the warm-start setting, the item has collected a small number of clicks and impressions. The model should use these interactions to adapt to the item while avoiding overfitting. Different items may depend on different interactions; for example, an expensive product may depend on user income and price, whereas another item may depend more on age and category.

After sufficient interactions are collected, the item enters the mature setting, where standard CTR models can learn stable item representations. The overall transition is from cold start to warm start and finally to the mature-item setting.

1.3 Problem Formulation

Let $\mathcal{V} = \{v_i\}$ be the item set and $\mathcal{U} = \{u_j\}$ be the user set. Each item and user is represented by their corresponding features. For a user–item pair (u_j, v_i), the binary label $y_{i,j} \in \{0, 1\}$ indicates whether the user clicks the item.

The goal is to learn a predictor f_θ that estimates the click probability as $\hat{y}_{i,j} = f_\theta(u_j, v_i) = P(y_{i,j} = 1 \mid u_j, v_i)$.

Each item defines a CTR prediction task $\mathcal{T}_i$ containing a support set $\mathcal{S}_i = \{(v_i, u_j, y_{i,j})\}_{j=1}^{N_s}$ and a query set $\mathcal{Q}_i = \{(v_i, u_j, y_{i,j})\}_{j=1}^{N_q}$. The support set is used for item-specific adaptation, while the query set is used for evaluation.

At test time, the model is applied to an unseen item v_k. The item is in the cold-start setting when $|\mathcal{S}_k| = 0$, in the warm-start setting when $0 < |\mathcal{S}_k| \ll N_{\text{mature}}$, and in the mature setting when $|\mathcal{S}_k| \geq N_{\text{mature}}$.

The model learns its parameters by minimizing the expected query-set loss, written as $\min_\theta \mathbb{E}_{\mathcal{T}_k}[\mathcal{L}(\mathcal{Q}_k; \theta, \mathcal{S}_k)]$. The main objective is to transfer knowledge from old items while learning item-specific feature interactions from limited new-item data.

2 Dataset and Evaluation Metrics

2.1 Dataset

We use the MovieLens-1M benchmark [Harper and Konstan, 2015], which contains approximately one million user–movie interactions. Movie features include movie ID, title, release year, and genres, while user features include user ID, age, gender, occupation, and zip code.

The original ratings are converted into binary labels, where $y_{i,j} = 0$ if $r_{i,j} < 4$ and $y_{i,j} = 1$ if $r_{i,j} \geq 4$. Therefore, ratings of at least 4 are treated as positive interactions.

MovieLens-1M was selected as the primary dataset because its moderate size and limited number of feature fields make the explicit construction of second- and third-order hyperedges computationally feasible. However, because MovieLens provides ratings rather than actual impression and click logs, it is used as a binary preference prediction benchmark that approximates the cold-start CTR prediction setting.

2.2 Evaluation Metrics

We evaluate the models using AUC and F1 score [Ling et al., 2003, Huang et al., 2015]. AUC measures the model's ability to rank positive user–item interactions above negative ones and is used to evaluate the overall ranking quality of CTR predictions. F1 score measures the balance between precision and recall and is used to evaluate binary classification performance. Higher AUC and F1 scores indicate better predictive performance.

3 Models

Graph-based models are appropriate because MovieLens naturally forms a user–item interaction graph, allowing the model to capture collaborative and higher-order relationships. Compared with matrix factorization, they are also better suited to the cold-start and warm-start settings because item-side features and interaction patterns can help represent new items with limited history.

This section introduces the baseline models and the proposed graph-based CTR model. All methods receive the same user–item instance (u_j, v_i) and output a click probability $\hat{y}_{i,j}$. Let $N' = N_v + N_u$ denote the total number of item-side and user-side features in one instance.

3.1 Baselines

LightGCN LightGCN is a graph collaborative-filtering baseline that simplifies graph convolution for recommendation by keeping only neighborhood aggregation and removing feature transformation and nonlinear activation [He et al., 2020]. It constructs a bipartite interaction graph whose nodes are users and items and whose edges are historical user–item interactions. For user u and item v, the layer-wise propagation can be written as

$$\mathbf{z}_u^{(l)} = \sum_{v \in \mathcal{N}(u)} \frac{1}{\sqrt{|\mathcal{N}(u)|\,|\mathcal{N}(v)|}} \mathbf{z}_v^{(l-1)}, \qquad \mathbf{z}_v^{(l)} = \sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{|\mathcal{N}(v)|\,|\mathcal{N}(u)|}} \mathbf{z}_u^{(l-1)},$$

where $\mathcal{N}(\cdot)$ denotes graph neighbors. The final representation is the weighted sum of embeddings from all layers,

$$\mathbf{z}_u = \sum_{l=0}^{L} \beta_l \mathbf{z}_u^{(l)}, \qquad \mathbf{z}_v = \sum_{l=0}^{L} \beta_l \mathbf{z}_v^{(l)},$$

and the CTR score is computed by $\hat{y}_{u,v} = \sigma(\mathbf{z}_u^\top \mathbf{z}_v)$. LightGCN is useful as a strong graph-based recommendation baseline because it captures high-order collaborative signals through repeated propagation on the user–item graph. However, it relies heavily on observed interactions. Therefore, for a newly introduced item with no or very few edges, the item embedding is difficult to estimate reliably.

EmerG EmerG is the main cold-start CTR baseline considered in this project [Wang et al., 2024]. Instead of learning a single global feature-interaction pattern, EmerG constructs an item-specific feature graph for each item. The nodes of this graph correspond to the N' user and item features, and edges represent feature interactions that are important for the target item.

Given item-side feature embeddings of item v_i, a hypernetwork generates the initial adjacency matrix $\bar{A}_i^{(1)}$ row by row:

$$[\bar{A}_i^{(1)}]_{m:} = \mathrm{MLP}_{W_a}\left([\mathbf{e}_{1,i}, \ldots, \mathbf{e}_{N_v,i}, \mathrm{onehot}(m)]\right),$$

where $[\bar{A}_i^{(1)}]_{m:}$ denotes the mth row and W_a is the hypernetwork parameter. Higher-order adjacency matrices are then generated by matrix powers:

$$\bar{A}_i^{(l)} = \bar{A}_i^{(l-1)} \bar{A}_i^{(1)}.$$

EmerG further normalizes the dense matrix, keeps only the top-K edge weights, symmetrizes the result, and applies a mask so that disconnected low-order edges remain disconnected in higher-order graphs. The resulting adjacency matrix $A_i^{(l)}$ is used in a customized message passing layer:

$$\mathbf{h}_{m,i}^{(l)} = \mathbf{h}_{m,i}^{(l-1)} \odot \left(\sum_{n=1}^{N'} [A_i^{(l)}]_{mn} W_g^{(l)} \mathbf{h}_{n,i}^{(0)} \right),$$

where $\odot$ is element-wise multiplication and $\mathbf{h}_{m,i}^{(0)} = \mathbf{e}_{m,i}$. By multiplying the previous feature representation with first-order feature embeddings, this message passing mechanism can capture high-order feature interactions. EmerG combines node representations from different layers using attention and predicts CTR through a feature-contribution network. It is trained with a meta-learning strategy that optimizes shared hypernetwork and GNN parameters across old-item tasks while updating only a small set of item-specific parameters during warm-up.

3.2 Proposed Method

We propose *Differentiable Graph Diffusion* (DGD), a compact extension of EmerG. DGD keeps EmerG's embedding layer, message passing rule, prediction head, and meta-learning loop, but replaces hard high-order graph construction with a differentiable diffusion module. EmerG has two key limitations: the lth layer uses only the exact power $(\bar{A}_i^{(1)})^l$, which cannot learn a weighted mixture of lower- and higher-order signals; and its top-K sparsification cuts gradients to discarded edges, making graph-structure learning less smooth.

For item v_i, DGD constructs the layer-l adjacency matrix as

$$A_i^{(l)} = \mathrm{sparsemax}\left(\sum_{k=1}^{l} \alpha_k^{(l)} \left(\bar{A}_i^{(1)}\right)^k \right), \qquad \alpha^{(l)} = [\alpha_1^{(l)}, \ldots, \alpha_l^{(l)}] \in \Delta^{l-1},$$

where $\alpha^{(l)}$ is a learnable simplex vector that controls the contribution of each graph power. Sparsemax is applied row-wise to produce sparse but differentiable neighbor distributions [Martins and Astudillo, 2016]; it can also be replaced by α-entmax for tunable sparsity [Peters et al., 2019]. DGD then uses the original EmerG message passing rule:

$$\mathbf{h}_{m,i}^{(l)} = \mathbf{h}_{m,i}^{(l-1)} \odot \left(\sum_{n=1}^{N'} [A_i^{(l)}]_{mn} W_g^{(l)} \mathbf{h}_{n,i}^{(0)} \right).$$

DGD therefore changes only graph generation, not feature propagation. It reduces to EmerG when $\alpha_k^{(l)} = \mathbb{1}[k = l]$ and sparsemax is replaced by hard top-K. Since the message passing rule is unchanged, EmerG's claim that the $(l-1)$th layer captures lth-order feature interactions is preserved, while DGD learns a soft mixture of orders 1 through l. Memory also remains $O(1)$ with respect to the number of layers because only $\bar{A}_i^{(1)}$ is stored per item.

We evaluate DGD using the same MovieLens and Taobao protocol as the baselines in Cold-Start, Warm-Up A, Warm-Up B, and Warm-Up C phases. Ablations include *G-Diffusion* + *Top-K* to test order mixing, *Power* + *Sparsemax* to test differentiable sparsification, and the full *G-Diffusion* + *Sparsemax* model to test their combined effect.

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